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Laser-Ablation Synthesis of Colloidal ZrN Nanoparticles in Different Liquids

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Abstract—The paper presents a study of the effect of the liquid medium type on the properties of zirconium nitride (ZrN) nanoparticles (NPs) synthesized by pulsed laser ablation in liquids (PLAL). A comparative analysis of the properties of NPs synthesized in six different liquids (acetonitrile, acetone, ethanol, isopropyl alcohol, dimethylformamide, and deionized water) demonstrates that NPs obtained in acetonitrile and acetone are the most promising for biomedical applications. Synthesized ZrN NPs can be used as sensitizers of innovative noninvasive methods of cancer therapy and contrast agents in diagnostics.

Keywords: zirconium nitride, transition metal nitrides, nanoparticles, laser ablation in liquid

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1. INTRODUCTION

The rapid development of nanotechnology has led to the emergence of many new promising therapeutic and diagnostic methods and materials. In particular, classical plasmonic nanomaterials, such as gold (Au) and silver (Ag) nanoparticles (NPs), have already demonstrated their effectiveness as sensitizers not only in traditional approaches to the treatment and diagnosis of malignant neoplasms, but also in new innovative methods of minimally invasive and noninvasive treatment, such as targeted photothermal therapy (PTT) and photoacoustic imaging (PAI) [1–3]. The PTT and PAI methods are based on a selective heating of NPs localized in the tumor by external light radiation. To this end, near-IR radiation with wavelengths in the range from 650 to 950 nm, in which the so-called first window of relative transparency of biological tissues is located is typically used. However, NPs of classical plasmonic materials (Au, Ag, and Cu), traditionally employed for the tasks of sensitization of PTT and contrasting in PAI, have their plasmonic absorption bands outside this range. In this regard, in real applications their absorption peaks can be shifted to the IR region by changing the shape of nanostructures (nanorods, core–shell nanoparticles, etc.) [4]. However, the use of such approaches still has a number of serious limitations in applicability.

A more promising way to reach high absorption in the transparency window of biological tissues may be the use of alternative plasmonic nanomaterials based on nitrides of transition metals of group IV (TiN, ZrN, and HfN) [5–7]. There are numerous experimental works confirming the high efficiency of TiN NPs in PTT and PAI tasks [5, 8]. At the same time, theoretical works predict that compared to TiN NPs, ZrN NPs should have more promising optical properties [9]. At the same time, ZrN NPs remain the least studied material in this class, and this work is intended to fill this gap.

Existing methods for the synthesis of ZrN NPs [6, 9] leave unresolved the problems of obtaining particles with suitable geometry, composition, purity, and good dispersibility in aqueous media for applications in biomedicine. In this context, pulsed laser ablation in liquids (PLAL) [10] is of particular interest. This synthesis method allows one, on the one hand, to overcome the limitations of chemical approaches associated with contamination of the NP surface by precursor residues and reaction by-products, leading

to toxic effects, and on the other hand, it offers significant simplicity and flexibility of synthesis, allowing the production of stable colloidal solutions of pure NPs with controlled physicochemical properties from virtually any material [8, 11–18].

In this paper, we present the results of a study of the dependence of the main characteristics (optical, dimensional, and morphological properties, as well as colloidal stability) of laser-synthesized ZrN NPs on the type of a liquid medium, where the ablation is perfomed.

2. MATERIALS AND METHODS

The ZrN-based NPs were synthesized using PLAL developed by us in previous works [19–21]. The pressed ZrN target (GoodFellow, USA, purity 99.999%) was located vertically inside a glass cell filled with 20 mL of liquid. Six different liquids were used, namely, acetonitrile, acetone, ethanol, isopropanol, dimethylformamide (DMF), and deionized water. Laser radiation, from a Yb:KGW laser TETA-20 (wavelength, 1030 nm; pulse duration, 270 fs; pulse energy, 20 μJ; and repetition rate, 200 kHz; Avesta, Russia), was directed through a system of mirrors into a galvanometric scanner LScan-10 (Ateko-TM, Russia), moving the radiation along the target surface at a speed of 4 m/s. The radiation was focused on the target surface by a flat-field (F-theta) 100 mm objective. The thickness of the liquid layer along the beam path was 3 mm. The duration of the synthesis for all types of liquids was 10 min. The resulting colloidal solutions were collected after the synthesis and studied without any additional post-processing steps.

Optical extinction spectra from colloidal solutions of ZrN NPs were recorded in the spectral range of 330–1100 nm using a ML122 spectrophotometer (SOL Instruments, Belarus). Hydrodynamic size distributions of NPs were measured by dynamic light scattering (DLS) using a Zetasizer ZS (Malvern Instruments, England). Glass cells with a 10 mm optical path were used to characterize the optical and hydrodynamic properties of the obtained NPs. The surface morphology of the NPs was analyzed using a MAIA 3 scanning electron microscope (SEM) (Tescan, Czech Republic) at an accelerating voltage of 30 kV. The size distributions of the synthesized NPs were constructed by analyzing SEM images in the ImageJ software environment.

3. RESULTS AND DISCUSSION

In this work, we developed and tested the synthesis of ZrN NPs in various types of liquids. This choice of liquids for synthesis was due to their wide prevalence among works devoted to the synthesis of nanomaterials by PLAL, as well as the different oxygen content in the chemical composition of each of the used solvents, which often affects the properties of particles formed during laser ablation [19]. The criteria for evaluating the NPs were the optical, dimensional, and morphological properties, as well as the colloidal stability of the obtained solutions. During the synthesis, the color of solution gradually evolved showing various shades of blue, which served as an indication of the formation of ZrN NPs (Fig. 1a). When synthesized ZrN NPs in acetonitrile, acetone, ethanol, isopropanol, and DMF, the colloidal solutions exhibited a clearly expressed plasmon peak with maxima at 634, 606, 592, 614, and 622 nm, respectively (Fig. 1b). However, the NPs obtained in deionized water did not have a plasmon peak, which most likely indicates the oxidation of the NPs during the synthesis. Oxidation of NPs is a negative effect from the point of view of biomedical applications, since it decreases in the efficiency of X-ray absorption [25, 26], which makes ZrN NPs synthesized in water ineffective for absorbing IR radiation and consequently for the use in PTT and PAI.

Thus, among all the studied samples, the NPs obtained in acetonitrile had the greatest overlap of the plasmon peak with the first transparency window of biological tissues. We associate this result with a decrease in NPs oxidation caused by the absence of oxygen atoms in the acetonitrile molecules (chemical formula CH_3CN). The sample of NPs obtained in acetonitrile also had the highest optical density, which indirectly indicates the highest productivity of the synthesis of ZrN NPs in this liquid.

The size of NPs determines not only their colloidal stability and optical properties, but also greatly affects the efficiency of cellular internalization and the duration of circulation in the bloodstream [17]. An optimal size (diameter) of NPs for in vivo applications is considered to be the range from 8 to 50–80 nm [24]. This is due to the fact that NPs smaller than 8 nm are quickly excreted from the body through the renal glomerular filtration system, and the efficiency of "catching" NPs from the bloodstream by the macrophage system and their subsequent excretion from the body rapidly increases with increasing size [25]. In addition, many mechanisms of cellular internalization become ineffective for NPs with sizes approaching 100 nm and greater.

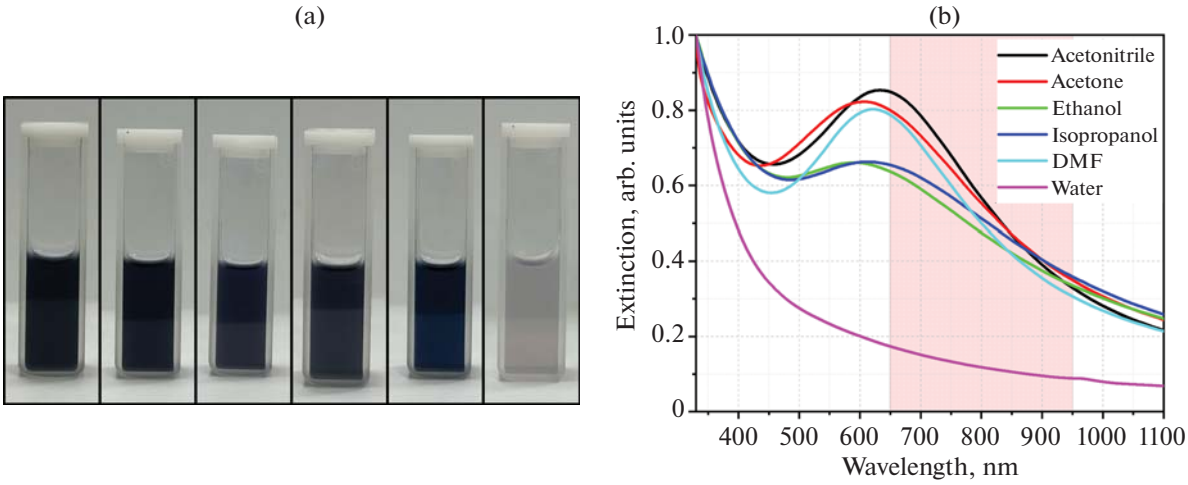

Fig. 1. Optical properties of ZrN NPs: (a) photographs of solutions (from left to right: acetonitrile, acetone, ethanol, isopropanol, DMF, and water); and (b) extinction spectra.

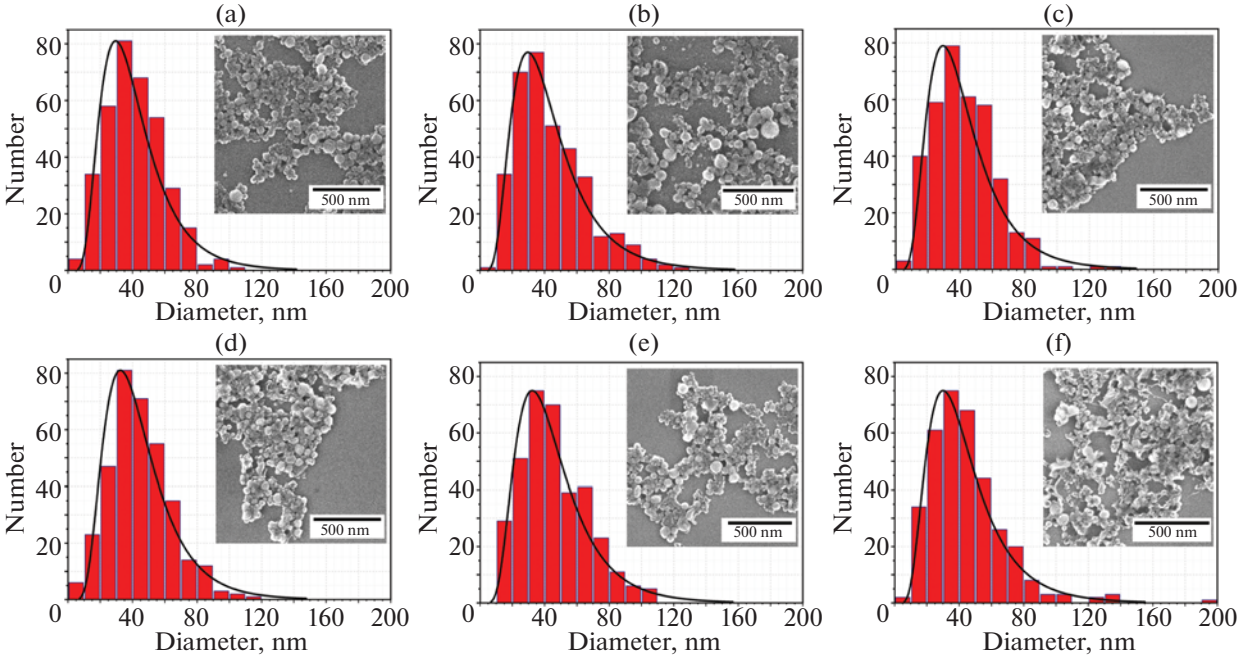

Fig. 2. Size distributions and images of ZrN NPs synthesized in (a) acetonitrile, (b) acetone, (c) ethanol, (d) isopropanol, (e) DMF, and (f) water.

Analysis of SEM images showed that the mode of the size distribution of ZrN NPs obtained in acetonitrile, acetone, ethanol, isopropanol, DMF, and deionized water was 29, 29, 29, 32, 32, and 30 nm, respectively (Fig. 2). Most of the NPs had a spherical shape regardless of the type of the liquid in which they were synthesized. However, the synthesis in alcohols and water led to the formation of a small (up to 10%) amount of NPs of irregular shape. Nevertheless, no significant differences in size and morphology were observed among all samples of the obtained NPs.

We also measured the hydrodynamic size distributions of ZrN NPs obtained by PLAL (Fig. 3). The peak of the hydrodynamic diameter distribution mode was located at 35, 44, 38, 30, 31, and 57 nm for NPs obtained in acetonitrile, acetone, ethanol, isopropanol, DMF, and water, respectively. The differences in the SEM data and the hydrodynamic diameter of NPs are due to the specifics of measurement

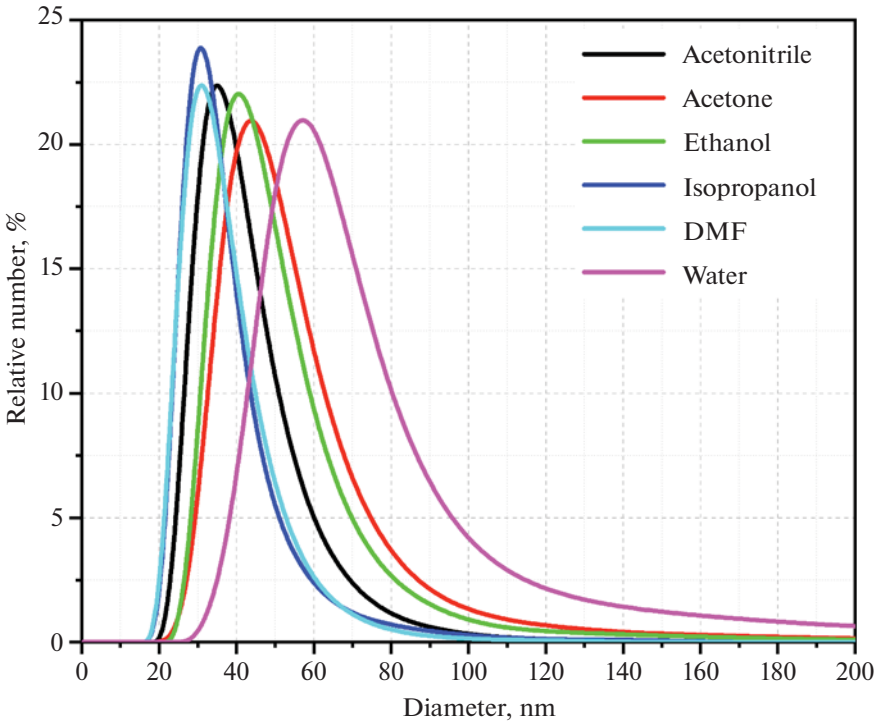

Fig. 3. Distribution of the hydrodynamic size of ZrN NPs.

by the DLS method, which does not allow small fractions of NPs to be taken into account against the background of larger ones.

In addition, we studied the colloidal stability of all obtained solutions. The NPs synthesized in acetonitrile, acetone, isopropanol, and deionized water had the highest stability. The NPs synthesized in ethanol began to agglomerate and precipitate to the bottom of the test tube within 30 min. The NPs synthesized in DMF precipitated within 4–5 days after synthesis. The remaining samples retained colloidal stability for at least several weeks.

Thus, ZrN NPs synthesized by PLAL in acetonitrile have the highest absorption in the transparency window of biological tissues among all the studied samples. At the same time, these NPs had the smallest size distribution mode (29 nm). The NPs obtained in acetonitrile and acetone show excellent colloidal stability; i.e., the particles do not agglomerate for at least 2 months after synthesis, which makes them extremely promising for biomedical applications as sensitizers of PTT and contrast agents in PAI.

4. CONCLUSIONS

Thus, we have studied optical, dimensional, and morphological properties of ZrN NPs synthesized in six different types of liquids. A comparative analysis of the properties of all samples has demonstrated that NPs obtained in acetonitrile and acetone have the greatest potential for their further use as sensitizers for PTT, as well as other noninvasive methods for diagnosing and treating cancer tumors.

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CONFLICT OF INTEREST

The authors of this work declare that they have no conflicts of interest.

REFERENCES

1. Grigalavicius, M., Mastrangelopoulou, M., Berg, K., Arous, D., Ménard, M., Raabe-Henriksen, T., Brondz, E., Siem, S., Görgen, A., Edin, N.F.J., Malinen, E., and Theodossiou, T.A., Proton-dynamic therapy following photosensitiser activation by accelerated protons demonstrated through fluorescence and singlet oxygen production, *Nat. Commun.,* 2019, vol. 10, p. 3986. https://doi.org/10.1038/s41467-019-12042-7
2. Liu, Y., Bhattarai, P., Dai, Z., and Chen, X., Photothermal therapy and photoacoustic imaging via nanotheranostics in fighting cancer, *Chem. Soc. Rev.,* 2019, vol. 48, pp. 2053–2108. https://doi.org/10.1039/C8CS00618K
3. Wust, P., Hildebrandt, B., Sreenivasa, G., Rau, B., Gellermann, J., Riess, H., Felix, R., and Schlag, P., Hyperthermia in combined treatment of cancer, *Lancet Oncol.,* 2002, vol. 3, pp. 487–497. https://doi.org/10.1016/S1470-2045(02)00818-5
4. Pérez-Juste, J., Pastoriza-Santos, I., Liz-Marzán, L.M., and Mulvaney, P., Gold nanorods: Synthesis, characterization and applications, *Coord. Chem. Rev.,* 2005, vol. 249, pp. 1870–1901. https://doi.org/10.1016/j.ccr.2005.01.030
5. He, W., Ai, K., Jiang, C., Li, Y., Song, X., and Lu, L., Plasmonic titanium nitride nanoparticles for in vivo photoacoustic tomography imaging and photothermal cancer therapy, *Biomaterials,* 2017, vol. 132, pp. 37–47. https://doi.org/10.1016/j.biomaterials.2017.04.007
6. Wang, H., Li, J., Li, K., Lin, Y., Chen, J., Gao, L., Nicolosi, V., and Xiao, X., Transition metal nitrides for electrochemical energy applications, *Chem. Soc. Rev.,* 2021, vol. 50, pp. 1354–1390. https://doi.org/10.1039/D0CS00415D
7. Biliak, K., Protsak, M., Pleskunov, P., Nikitin, D., Hanuš, J., Ali-Ogly, S., Šomvársky, J., Tosca, M., Cieslar, M., Košutová, T., Dopita, M., Lopes Ferreira, F., and Choukourov, A., Plasmonic TiN, ZrN, and HfN nanofluids for solar-to-heat conversion, *ACS Appl. Nano Mater.,* 2023, vol. 6, pp. 21642–21651. https://doi.org/10.1021/acsanm.3c03662
8. Popov, A.A., Tselikov, G., Dumas, N., Berard, C., Metwally, K., Jones, N., Al-Kattan, A., Larrat, B., Braguer, D., Mensah, S., Da Silva, A., Estève, M.-A., and Kabashin, A.V., Laser-synthesized TiN nanoparticles as promising plasmonic alternative for biomedical applications, *Sci. Rep.,* 2019, vol. 9, p. 1194. . https://doi.org/10.1038/s41598-018-37519-1
9. Lalisse, A., Tessier, G., Plain, J., and Baffou, G., Plasmonic efficiencies of nanoparticles made of metal nitrides (TiN, ZrN) compared with gold, *Sci. Rep.,* 2016, vol. 6, p. 38647. https://doi.org/10.1038/srep38647
10. Fojtik, A. and Henglein, A., Laser ablation of films and suspended particles in a solvent: Formation of cluster and colloid solutions, *Ber. Bunsenges. Phys. Chem.,* 1993, vol. 97, p. 252.
11. Bulmahn, J.C., Tikhonowski, G., Popov, A.A., Kuzmin, A., Klimentov, S.M., Kabashin, A.V., and Prasad, P.N., Laser-ablative synthesis of stable aqueous solutions of elemental bismuth nanoparticles for multimodal theranostic applications, *Nanomaterials,* 2020, vol. 10, p. 1463. https://doi.org/10.3390/nano10081463
12. Al-Kattan, A., Tselikov, G., Metwally, K., Popov, A.A., Mensah, S., and Kabashin, A.V., Laser ablation-assisted synthesis of plasmonic Si@Au core-satellite nanocomposites for biomedical applications, *Nanomaterials,* 2021, vol. 11, p. 592. https://doi.org/10.3390/nano11030592
13. Pastukhov, A.I., Belyaev, I.B., Bulmahn, J.C., Zelepukin, I.V., Popov, A.A., Zavestovskaya, I.N., Klimentov, S.M., Deyev, S.M., Prasad, P.N., and Kabashin, A.V., Laser-ablative aqueous synthesis and characterization of elemental boron nanoparticles for biomedical applications, *Sci. Rep.,* 2022, vol. 12, p. 9129. https://doi.org/10.1038/s41598-022-13066-8
14. Griaznova, O.Yu., Belyaev, I.B., Sogomonyan, A.S., Zelepukin, I.V., Tikhonowski, G.V., Popov, A.A., Komlev, A.S., Nikitin, P.I., Gorin, D.A., and Kabashin, A.V., Laser-synthesized core-satellite Fe-Au nanoparticles for multimodal in vivo imaging and in vitro photothermal therapy, *Pharmaceutics,* 2022, vol. 14, p. 994. https://doi.org/10.3390/pharmaceutics14050994
15. Popov, A.A., Swiatkowska-Warkocka, Z., Marszalek, M., Tselikov, G., Zelepukin, I.V., Al-Kattan, A., Deyev, S.M., Klimentov, S.M., Itina, T.E., and Kabashin, A.V., Laser-ablative synthesis of ultrapure magneto-plasmonic core-satellite nanocomposites for biomedical applications, *Nanomaterials,* 2022, vol. 12, p. 649. https://doi.org/10.3390/nano12040649
16. Bailly, A.-L., Correard, F., Popov, A., Tselikov, G., Chaspoul, F., Appay, R., Al-Kattan, A., Kabashin, A.V., Braguer, D., and Estève, M.-A., In vivo evaluation of safety, biodistribution and pharmacokinetics of laser-synthesized gold nanoparticles, *Sci. Rep.,* 2019, vol. 9, p. 12890. https://doi.org/10.1038/s41598-019-48748-3
17. Zelepukin, I.V., Popov, A.A., Shipunova, V.O., Tikhonowski, G.V., Mirkasymov, A.B., Popova-Kuznetsova, E.A., Klimentov, S.M., and Kabashin, A.V., Laser-synthesized TiN nanoparticles for biomedical applications: Evaluation of safety, biodistribution, and pharmacokinetics, *Mater. Sci. Eng. C,* 2021, vol. 120, p. 111717. https://doi.org/10.1016/j.msec.2020.111717

18. Tselikov, G.I., Ermolaev, G.A., Popov, A.A., Tikhonowski, G.V., Panova, D.A., Taradin, A.S., Vyshnevyy, A.A., Syuy, A.V., Klimentov, S.M., Novikov, S.M., Evlyukhin, A.B., Kabashin, A.V., Arsenin, A.V., Novoselov, K.S., and Volkov, V.S., Transition metal dichalcogenide nanospheres for high-refractive-index nanophotonics and biomedical theranostics, *PNAS,* 2022, vol. 119. https://doi.org/10.1073/pnas.2208830119
19. Popov, A.A., Tikhonowski, G.V., Shakhov, P.V., Popova-Kuznetsova, E.A., Tselikov, G.I., Romanov, R.I., Markeev, A.M., Klimentov, S.M., and Kabashin, A.V., Synthesis of titanium nitride nanoparticles by pulsed laser ablation in different aqueous and organic solutions, *Nanomaterials,* 2022, vol. 12, p. 1672. https://doi.org/10.3390/nano12101672
20. Savinov, M.S., Griaznova, O.Y., Tikhonowski, G.V., et al., X-ray contrast properties of bismuth-based nanoformulations, *Bull. Leb. Phys. Inst. Suppl. QO,* 2023, vol. 50, no. 11, p. S1265. https://doi.org/10.3103/S1068335623220141
21. Zavestovskaya, I.N., Popov, A.L., Kolmanovich, D.D., Tikhonowski, G.V., Pastukhov, A.I., Savinov, M.S., Shakhov, P.V., Babkova, J.S., Popov, A.A., Zelepukin, I.V., Grigoryeva, M.S., Shemyakov, A.E., Klimentov, S.M., Ryabov, V.A., Prasad, P.N., Deyev, S.M., and Kabashin, A.V., Boron nanoparticle-enhanced proton therapy for cancer treatment, *Nanomaterials,* 2023, vol. 13, p. 2167. https://doi.org/10.3390/nano13152167
22. Song, G., Cheng, L., Chao, Y., Yang, K., and Liu, Z., Emerging nanotechnology and advanced materials for cancer radiation therapy, *Adv. Mater.,* 2017, vol. 29, p. 1700996. https://doi.org/10.1002/adma.201700996
23. De La Vega, J.C. and Häfeli, U.O., Utilization of nanoparticles as X-ray contrast agents for diagnostic imaging applications, *Contrast Media Mol. Imaging,* 2015, vol. 10, pp. 81–95. https://doi.org/10.1002/cmmi.1613
24. Hoshyar, N., Gray, S., Han, H., and Bao, G., The effect of nanoparticle size on in vivo pharmacokinetics and cellular interaction, *Nanomedicine,* 2016, vol. 11, pp. 673–692. https://doi.org/10.2217/nnm.16.5
25. Petros, R.A. and DeSimone, J.M., Strategies in the design of nanoparticles for therapeutic applications, *Nat. Rev. Drug Discov.,* 2010, vol. 9, pp. 615–627. https://doi.org/10.1038/nrd2591

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